Casey D. Foley, PhD

Laser and Vacuum Systems | R&D | Production Optimization | Instrumentation | Machine Vision

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KEY ACHIEVEMENTS

- **99% laser process yield** achieved independently across two separate laser ablation systems through sensor integration, process design, and statistical process control
- **3x manufacturing takt time reduction** achieved across multiple systems through laser system upgrades, parameter optimization, and Python-based automation
- **2x throughput increase** in vacuum insulated glass evacuation process with 99% yield and improved unit R-Value
- **$1M+ instrument recovery** — full restoration of cryogenic ion trap including UHV rebuild, laser integration, and instrument control programming
- **Built root cause analysis platform**, improving yield and enabling company-wide data communication
- **20+ peer-reviewed publications** including *Science* and *JACS*; **430+ citations**; **15+ conference presentations**

CORE COMPETENCIES

Laser Systems & Process Engineering
Fiber • Nd:YAG • DPSS • CO_2 • Excimer • VUV • IR OPO/OPA • Ti:Sapphire • Dye • Ablation • Scribe • Heating • Process Development • Optical Alignment • Beam Diagnostics

Advanced Instrumentation & Vacuum
Ultra-High Vacuum Systems • Machine Vision • Sensor Integration • Mass Spectrometry (Waters, Thermo, Bruker) • Ion Mobility • Spectroscopy (UV/Vis/IR/FTIR/REMPI) • Particle Imaging • SolidWorks CAD

Programming & Data Science
Python • MATLAB • LabVIEW • Machine Learning • Data Analytics • Automation Scripting • Data Acquisition Systems • Statistical Analysis

Manufacturing & Quality Systems
SPC • Cpk/Ppk • PFMEA • Root Cause Analysis • PDCA • DOE • SOP & Control Plan Authoring • Preventative Maintenance • Yield Optimization • KPI Development

PROFESSIONAL EXPERIENCE

Laser Systems Engineer
August 2024 – Present
LuxWall

- **Integrated fiber laser ablation system**, achieving 99% yield and 3× takt time reduction; separately achieved 99% yield and 2x takt time reduction on a second ablation process through sensor integration and SPC
- **Led development of vacuum-insulated glass evacuation process** in first factory, achieving 2× throughput increase, 99% yield, and improved unit R-Value
- **Led development of laser heating processes** for vacuum-insulated glass production
- **Served as cross-functional technical liaison** between factory engineering,Scaling, and R&D teams
- **Machine vision and sensor integration for quality control** — projects include single glass pane defect detection system deployment, material color measurement integration, and ML-based surface inspection
- **SME for all laser and high vacuum processes** — authored SOPs, work instructions, control plans, and troubleshooting guides; overhauled preventative maintenance procedures

Research Excellence Postdoctoral Fellow
2022 – July 2024
University of Missouri

- **Designed and built custom UHV imaging instrument** for hypersonic collision studies with electron-ion coincidence detection
- **Implemented laser system for novel VUV detection** of atoms and molecules; performed MeV-UED experiments at SLAC National Laboratory using advanced instrumentation and custom Python analysis
- **Mentored and trained research group members** as senior postdoc; led lab operations and instrument training

Postdoctoral Associate
2021 – 2022

Sandia National Laboratories

- **Restored \$1M+ cryogenically-cooled ion trap instrument** including UHV system rebuild, IR OPO/OPA laser installation, and programming of LabVIEW control and data analysis software
- **Developed IR, UV, and double resonance spectroscopy methods** for novel molecular detection and structural characterization of metal–molecule complexes

Graduate Research Assistant
May 2016 – 2020

University of Missouri

- **Published breakthrough research in *Science*** on quantum resonances in molecular dynamics using velocity map imaging
- **Designed comprehensive laser automation systems** and custom optical detection for UHV molecular beam experiments

Graduate Research Assistant
2014 – April 2016

Wayne State University

- **Developed mass spectrometry methods** across Waters, Thermo, and Bruker platforms with ion mobility for biological and polymer analysis
- **Taught analytical chemistry laboratory courses** including chromatography and separation science

Summer Journeyman Fellow
Summer 2018

Army Research Laboratory

- **Developed porous silicon particle workflow** from wafer processing to particle fractionation for energetic materials; characterized via FTIR, calorimetry, dynamic pressure measurements, and porosimetry

EDUCATION

PhD, Physical Chemistry
December 2020

University of Missouri — Dissertation: Quantum Aspects of Roaming Dynamics — Advisor: Prof. Arthur G. Suits

BS, Biochemistry
May 2013

Saginaw Valley State University — President's Scholarship — Industrial experience at Dow Chemical

SELECTED PUBLICATIONS *(complete list: 20+ publications, 430+ citations)*

- **Site-specific Photochemistry along a Protonated Peptide Scaffold** — *JACS, 146: 13282–13295, 2024*
- **Orbiting Resonances in Formaldehyde Reveal Coupling of Roaming, Radical, and Molecular Channels** — *Science, 374: 1122–1127, 2021*
- **Molecular Cage Reports on Its Contents: Spectroscopic Signatures of Cryo-Cooled K^+- and Ba^{2+}-Benzocryptand Complexes** — *J. Phys. Chem. A, 127: 6227–6240, 2023*
- **Ion Mobility Spectrometry-Mass Spectrometry to Elucidate Arm-Dispersity within Star Polymers** — *ACS MacroLetters, 4: 778–782, 2015*

HONORS & AWARDS

- Army Research Lab Journeyman Fellowship (2018)
- WSU Thomas C. Rumble Fellowship (2014–2015)
- WSU Chemistry Citation for Excellence in Teaching (2014)
- SVSU President's Scholarship (2009–2013)